

XIAOYU LI

Nanjing University

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BIOGRAPHY

I am a Ph.D. student in the School of Computer Science at Nanjing University, under the supervision of Dr. Jia Liu. Previously, I received my B.E. degree from Huazhong University of Science and Technology (HUST). My primary research interests focus on mobile computing, wireless communication, and AI-driven wireless sensing.

EDUCATION

Nanjing University

Sep 2023 – June 2028 (Expected)

Ph.D. Student in the School of Computer Science

Supervisor: Dr. Jia Liu, Associate Professor

Huazhong University of Science and Technology

2019 – 2023

B.Eng. in the School of Computer Science and Technology

AWARDS AND HONORS

- National Scholarship, China *2024*
- Gold Medal in the 49th Geneva's Invention Expo, WIPO *2024*
- First Prize in the Ubiquitous Intelligent Sensing Technology Innovation and Application Competition, China *2024*

PUBLICATION

Accepted Papers:

- **[TMC'25]** Xiaoyu Li, Jia Liu, Zihao Lin, Xuan Liu, Yanyan Wang, Shigeng Zhang, Baoliu Ye, "Advancing RFID Technology for Virtual Boundary Detection", *IEEE Transactions on Mobile Computing*, 2025.
- **[INFOCOM'24]** Xiaoyu Li, Jia Liu, Xuan Liu, Yanyan Wang, Shigeng Zhang, Baoliu Ye, Lijun Chen, "RF-Boundary: RFID-Based Virtual Boundary", *The IEEE International Conference on Computer Communications*, 2024.
- **[INFOCOM'26]** Yuyao Zhong, Jia Liu, Xiaoyu Li, Xingyu Chen, Yanyan Wang, Zuojian Zhou, "Fast K-Reads in COTS RFID Systems", *The IEEE International Conference on Computer Communications*, 2026.
- **[TMC'26]** Chengxuan Fu, Jia Liu, Xiaoyu Li, Xingyu Chen, Xuan Liu, Shigeng Zhang, Qiguo Huang, Junzhao Du, "Maximizing RFID Coverage Capacity: From Theory to Practice", *IEEE Transactions on Mobile Computing*, 2026.
- **[INFOCOM'25]** Chengxuan Fu, Jia Liu, Xiaoyu Li, Xingyu Chen, Xuan Liu, Shigeng Zhang, Junzhao Du, "Exploring the Frontiers of RFID Coverage Capacity: Theoretical and Practical Perspectives", *The IEEE International Conference on Computer Communications*, 2025.

- [\[IWQoS'25\]](#) Xingyu Chen, Jia Liu, Haoyu Hu, Shaochuang Li, **Xiaoyu Li**, Zihao Lin, "RF-SLAM: Fusing RFID and LiDAR SLAM for Item-Level Semantic Mapping", *The IEEE/ACM International Symposium on Quality of Service, 2025*.

Manuscripts Under Review:

- [\[MobiCom'26\]](#) **Xiaoyu Li**, Jia Liu, Rui Shi, Haipeng Dai, Baoliu Ye, Guihai Chen, "GRaF: 3D Gaussian Ray Tracing for Radio-Frequency Radiance Fields", *The 32th Annual International Conference on Mobile Computing and Networking, 2026*. (One-Shot-Revision)
- [\[MobiSys'26\]](#) **Xiaoyu Li**, Jia Liu, Rui Shi, He Huang, Baoliu Ye, Junzhao Du, Guihai Chen, "LoT: Generalizable RF Localization with Transformers", *The 24th ACM International Conference on Mobile Systems, Applications, and Services, 2026*. (2nd-Round Revision)

RESEARCH PROJECTS

1. RFID-Enabled Intelligent Sensing.

This research focuses on enabling intelligent sensing by constructing an RFID-based virtual boundary through collaborative interaction among passive RFID nodes. It opens up a new avenue for building a battery-free, low-cost, easy-to-maintain virtual boundary, without the need for any system calibration. The main contributions include:

- RF-Boundary: RFID-Based Virtual Boundary
- Advancing RFID Technology for Virtual Boundary Detection

2. AI-Driven Wireless Sensing and Localization.

This research aims to model complex radio-frequency interactions by developing capabilities for RF field understanding and context sensing through carefully designed AI-driven models. These models are trained and evaluated on real-world datasets spanning diverse scenes and multiple applications, ranging from wireless channel estimation to localization. The main contributions include:

- GRaF: 3D Gaussian Ray Tracing for Radio-Frequency Radiance Fields
- LoT: Generalizable RF Localization with Transformers

3. Radio-Frequency Communication Optimization.

This research investigates how to improve the coverage capacity and reading reliability of commercial RFID systems through principled communication optimization. It aims to push the performance limits of large-scale RFID deployments by jointly analyzing physical-layer constraints and protocol-level inefficiencies. The main contributions include:

- Exploring the Frontiers of RFID Coverage Capacity: Theoretical and Practical Perspectives
- Maximizing RFID Coverage Capacity: From Theory to Practice
- Fast K-Reads in COTS RFID Systems

SERVICE

Reviewer:

- The 21st International Conference on Mobility, Sensing and Networking (2025)
- IEEE Transactions on Mobile Computing (2024)

TEACHING EXPERIENCE

Data Structure, Teaching Assistant. (Lecturer: Dr. Jia Liu.)

2023 – 2024